

Industrial Security (Bachelor) – Acronyms

OT, IT, security, and standards acronyms used across the course. Keep this sheet open during exercises and labs.

A

AD – Active Directory. The Windows domain service that holds enterprise identities; OT borrows it for engineering workstations.

ALARP – As Low As Reasonably Practicable. Safety-engineering principle imported into OT risk acceptance.

ANSI – American National Standards Institute. Co-publishes ISA standards (e.g. ANSI/ISA-84).

AP – Access Point (wireless).

APT – Advanced Persistent Threat. Long-dwell, well-resourced adversary; nation-state in practice.

AS – Application Server.

ATT&CK – MITRE's adversary tactics, techniques, and common knowledge matrix. ICS variant exists since 2020.

AV – Antivirus.

B

BCP – Business Continuity Plan.

BPCS – Basic Process Control System. The day-to-day DCS/PLC loop, as opposed to the SIS.

BSI – Bundesamt für Sicherheit in der Informationstechnik. German federal cyber-security authority.

BYOD – Bring Your Own Device. Generally forbidden in OT.

C

C2 – Command and Control (channel used by malware).

C2M2 – Cybersecurity Capability Maturity Model. US DOE maturity framework, widely used in OT.

CCE – Consequence-driven Cyber-informed Engineering. INL methodology, see Bochman & Freeman 2021.

CERT – Computer Emergency Response Team.

CIA – Confidentiality, Integrity, Availability. Re-ordered to A>I>C in OT.

CIP – 1. Critical Infrastructure Protection (NERC CIP-002–014, US electric). 2. Common Industrial Protocol (Rockwell, on EtherNet/IP).

CIRCIA – Cyber Incident Reporting for Critical Infrastructure Act (US, 2022).

CIS – Center for Internet Security; publishes hardening benchmarks.

CISA – Cybersecurity & Infrastructure Security Agency (US).

CISO – Chief Information Security Officer.

CMDB – Configuration Management Database (asset inventory of record).

CPU – Central Processing Unit.

CRA – Cyber Resilience Act – Regulation (EU) 2024/2847. Most provisions enter force 11 Dec 2027.

CSF – NIST Cybersecurity Framework. CSF 2.0 published Feb 2024 added the *Govern* function.

CSIRT – Computer Security Incident Response Team.

CTD – Continuous Threat Detection (Claroty).

CVE – Common Vulnerabilities and Exposures (MITRE-assigned ID).

CVSS – Common Vulnerability Scoring System. v3.1 most common in OT advisories.

CWE – Common Weakness Enumeration. CWE-798 (hard-

coded credentials) is the most-cited OT root cause.

D

DCS – Distributed Control System. Process-industry sibling of PLC + SCADA, typically vendor-integrated (Emerson, Honeywell, Yokogawa, ABB).

DMZ – Demilitarised Zone. In OT, the segment between IT and the plant (Purdue Level 3.5 / iDMZ).

DNP3 – Distributed Network Protocol 3. Common in North American utilities; SAV5 adds authentication, rarely enabled.

DoS – Denial of Service.

DSGVO – Datenschutz-Grundverordnung – the German name for the EU GDPR.

E

EDR – Endpoint Detection and Response.

ENISA – European Union Agency for Cybersecurity.

ESET – Slovak AV/security vendor; published the Industroyer 2017 analysis.

EU – European Union.

F

FAIR – Factor Analysis of Information Risk. Quantitative risk model by Jones & Freund.

FAT – Factory Acceptance Test.

FBI – Federal Bureau of Investigation.

FIRST – Forum of Incident Response and Security Teams; publishes CVSS and the PSIRT services framework.

FW – Firewall.

G

GDPR – General Data Protection Regulation (EU 2016/679).

GOOSE – Generic Object-Oriented Substation Event – IEC 61850 Layer-2 messaging, no auth by default.

GRU – Russian military intelligence; Unit 74455 = Sandworm.

H

HAZOP – HAZard and OPerability study; IEC 61882. *Cyber-HAZOP* adds an attacker overlay.

HMI – Human-Machine Interface (operator console).

HSE – Health, Safety, Environment.

I

IACS – Industrial Automation and Control Systems. The 62443 term for OT.

ICS – Industrial Control System.

ICS-CERT – Predecessor of CISA's OT advisory arm; advisories carry the ICSA- prefix.

ID – Identifier.

IDS – Intrusion Detection System.

IEC – International Electrotechnical Commission. Publishes 61131, 61850, 62443, 61508, 61511.

IED – Intelligent Electronic Device. Substation-side relay/controller.

IEEE – Institute of Electrical and Electronics Engineers. Publishes 802.1X, 1588.

IIoT – Industrial Internet of Things.

INL – Idaho National Laboratory. Author of CCE and the 2016 Defense-in-Depth recommended practice.

IP – Internet Protocol.

IPS – Intrusion Prevention System.

IR – Incident Response.

IRP – Incident Response Plan.

ISA – International Society of Automation. Co-publishes 62443 with IEC.

ISAC – Information Sharing and Analysis Center (e.g. E-ISAC for electric, WaterISAC).

ISIS – Institute for Science and International Security. Source of the canonical Natanz centrifuge estimate.

ISMS – Information Security Management System (ISO/IEC 27001).

ISO – International Organization for Standardization.

IT – Information Technology.

K

KPI – Key Performance Indicator.

KRITIS – German term for critical infrastructure; regulated under BSI-Kritisverordnung.

L

LAN – Local Area Network.

LOLBin – Living-Off-the-Land Binary – a legitimate OS tool repurposed by attackers (PowerShell, WMI, certutil).

LOPA – Layers Of Protection Analysis. AIChE CCPS quantitative safety method.

M

MAC – Media Access Control (address).

MBAP – Modbus Application Protocol header. 7 bytes prepended in Modbus TCP.

MES – Manufacturing Execution System (Purdue Level 3).

MFA – Multi-Factor Authentication.

MITRE – Non-profit that maintains CVE, CWE, ATT&CK.

MQTT – Message Queuing Telemetry Transport. Lightweight IIoT pub/sub; TLS optional (port 8883).

N

NAC – Network Access Control (e.g. 802.1X, Cisco ISE, Forescout).

NAT – Network Address Translation.

NERC – North American Electric Reliability Corporation; publishes the CIP standards.

NIS2 – EU Network and Information Systems Directive 2 – Directive (EU) 2022/2555. Transposition deadline 17 Oct 2024.

NIST – US National Institute of Standards and Technology.

NTP – Network Time Protocol.

NVD – NIST National Vulnerability Database.

O

OPC – Open Platform Communications. Vendor-neutral interoperability spec; OPC UA is the modern version.

OS – Operating System.

OT – Operational Technology. Hardware and software that interact with the physical world.

P

PCAP – Packet capture file (libpcap format).

PFD – Probability of Failure on Demand. Used in LOPA / SIL

credit.

PHA – Process Hazard Analysis. *Cyber-PHA* adds attacker-induced deviations.

PLC – Programmable Logic Controller.

PSIRT – Product Security Incident Response Team (vendor side).

PTP – Precision Time Protocol – IEEE 1588; sub-microsecond sync in substations.

R

RACI – Responsible, Accountable, Consulted, Informed (decision matrix).

RADIUS – Remote Authentication Dial-In User Service. Backs 802.1X NAC.

RBAC – Role-Based Access Control.

RPO – Recovery Point Objective. “How much data can we lose?”

RTO – Recovery Time Objective. “How long can we be down?”

RTU – Remote Terminal Unit. Field-side PLC equivalent in SCADA setups.

S

SCADA – Supervisory Control and Data Acquisition. The operator-side software watching a fleet of PLCs/RTUs.

SBOM – Software Bill of Materials. Required for CRA-scoped products.

SIEM – Security Information and Event Management.

SIL – Safety Integrity Level (IEC 61508/61511). SIL 1–4.

SIS – Safety Instrumented System. Independent layer that trips the process to a safe state.

SL – Security Level. IEC 62443 attacker tiers SL 1–4 (casual → nation-state).

SL-T – Target Security Level (assigned per zone).

SOC – Security Operations Center.

SP – Special Publication (NIST series, e.g. SP 800-82 R3).

SPAN – Switched Port Analyzer (Cisco term for port mirror; used to feed IDS).

SR – System Requirement (IEC 62443-3-3).

ST – Structured Text (IEC 61131-3 PLC language).

T

TAP – Test Access Point (dedicated tap device, more reliable than SPAN).

TCP – Transmission Control Protocol.

TDoS – Telephony Denial of Service.

TIA – Totally Integrated Automation (Siemens engineering toolchain).

TLS – Transport Layer Security.

TR – Technical Report (ISA, IEC).

TRISIS – Alternative name for TRITON / HatMan (Schneider Triconex SIS attack, 2017).

TRITON – Malware family targeting Schneider Triconex Safety Instrumented Systems (2017).

TSA – US Transportation Security Administration; issues Security Directives for pipelines and rail.

U

UA – Unified Architecture (the “UA” in OPC UA).

UDP – User Datagram Protocol.

UI – User Interface.

USB – Universal Serial Bus. Stuxnet vector; treat as untrusted-by-default in OT.

V

VFD – Variable Frequency Drive. Stuxnet’s eventual target.

VLAN – Virtual LAN.

VPN – Virtual Private Network.

W

WAN – Wide Area Network.

WPA – Wi-Fi Protected Access (WPA2-Enterprise / WPA3-Enterprise on OT Wi-Fi).

WS – Workstation. “Engineering WS” = engineer’s laptop with vendor toolchain.

Z

Zeek – Open-source network monitor / scriptable IDS (formerly Bro).